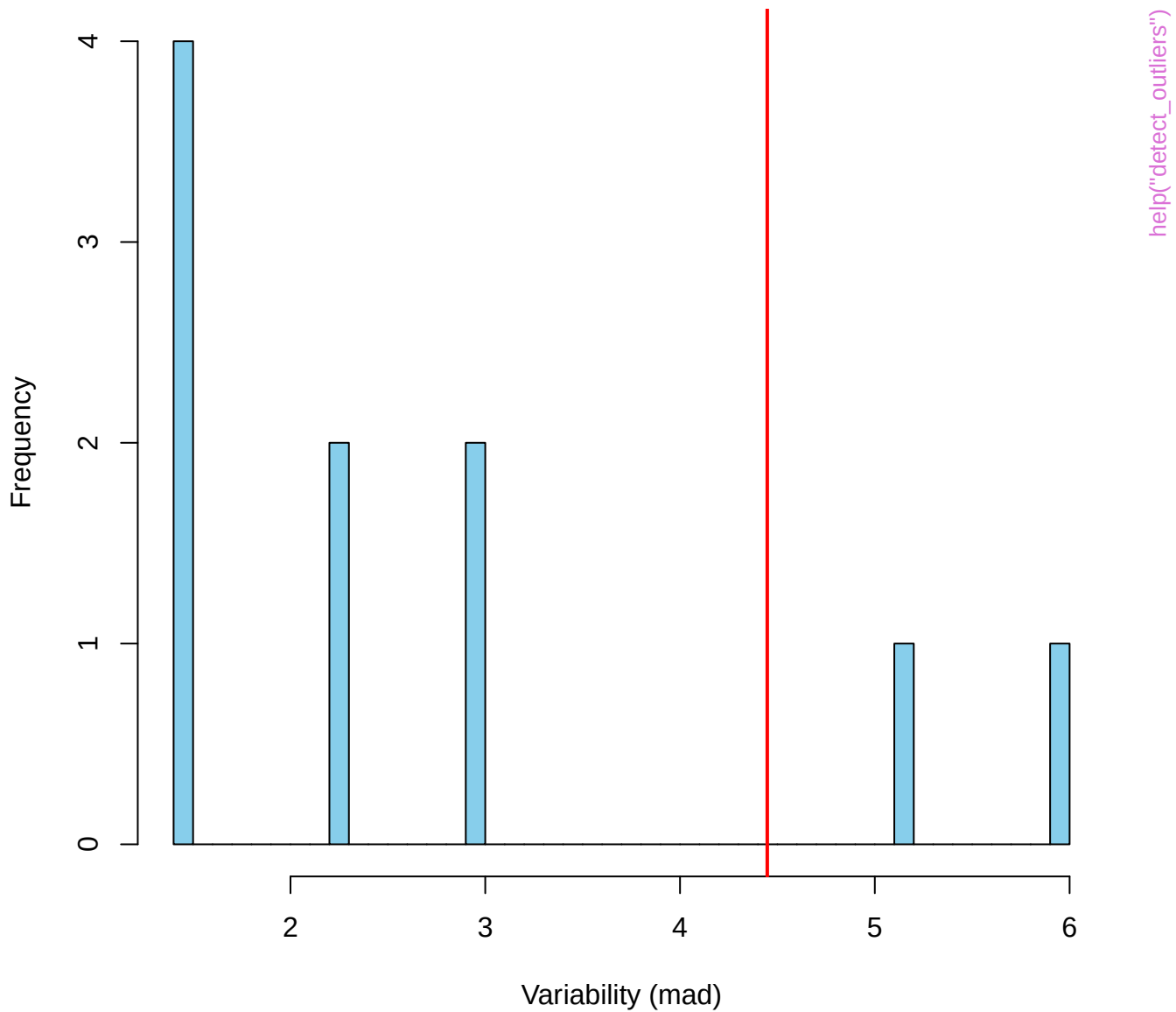
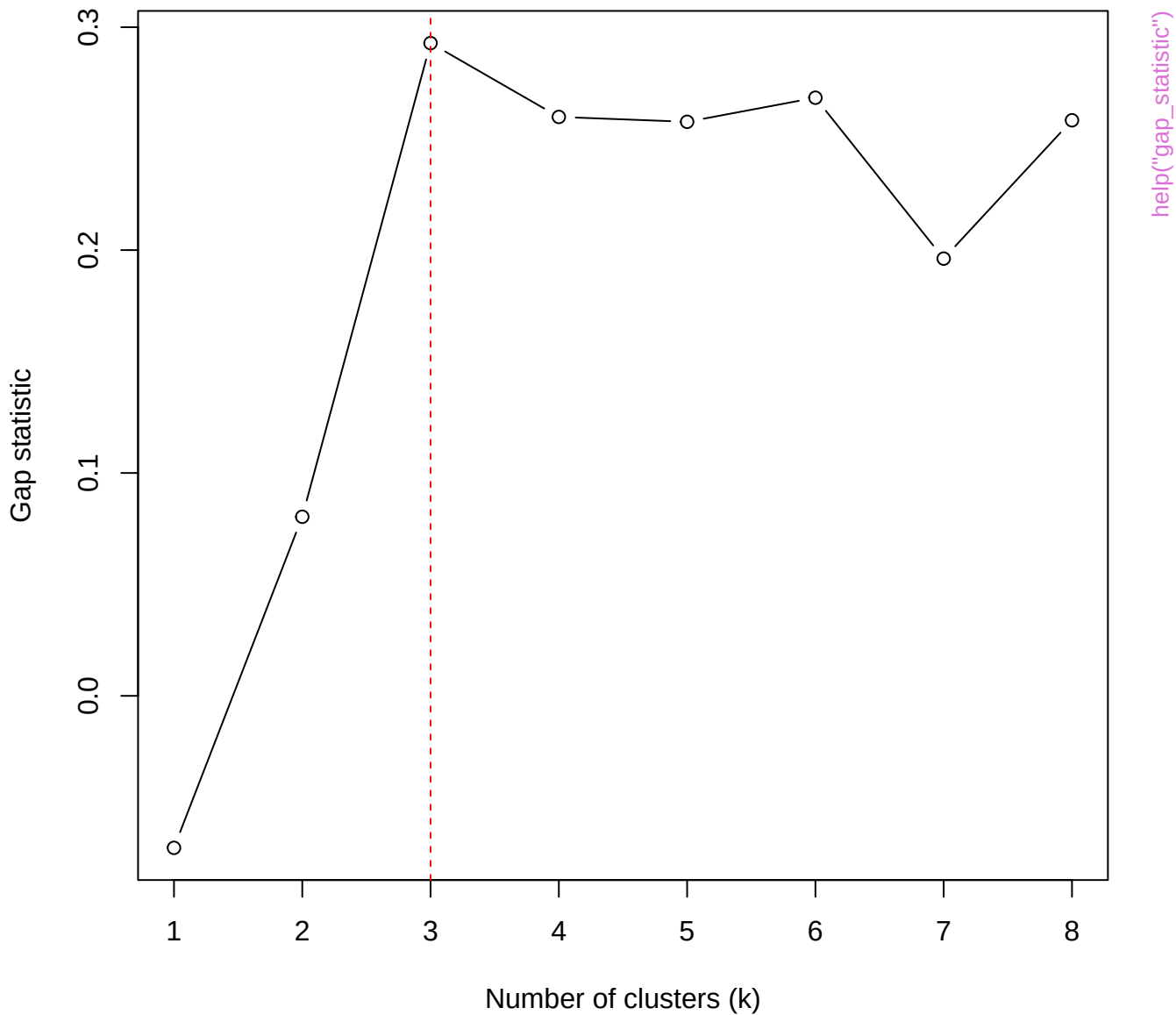


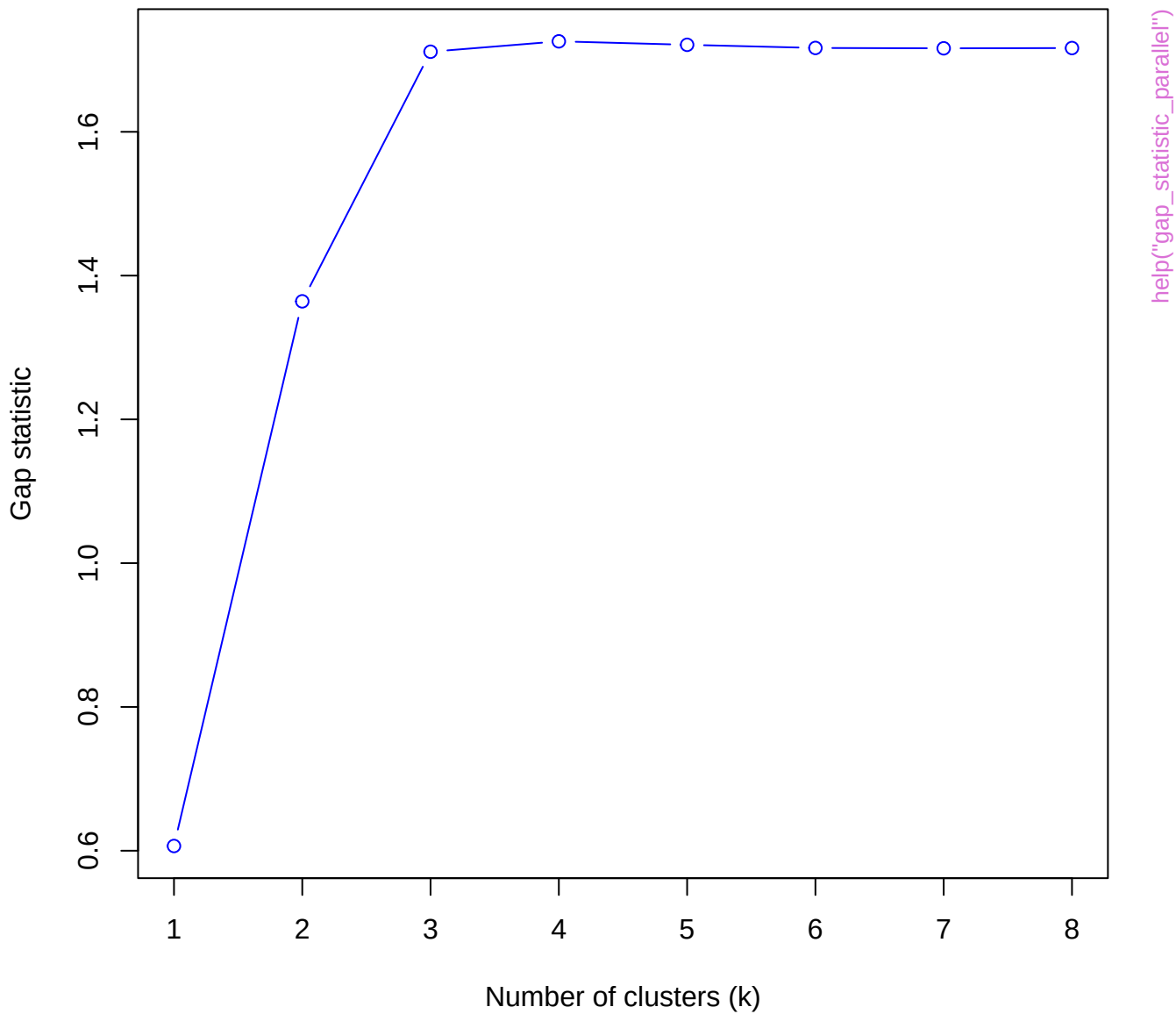
Gene Variability (mad)



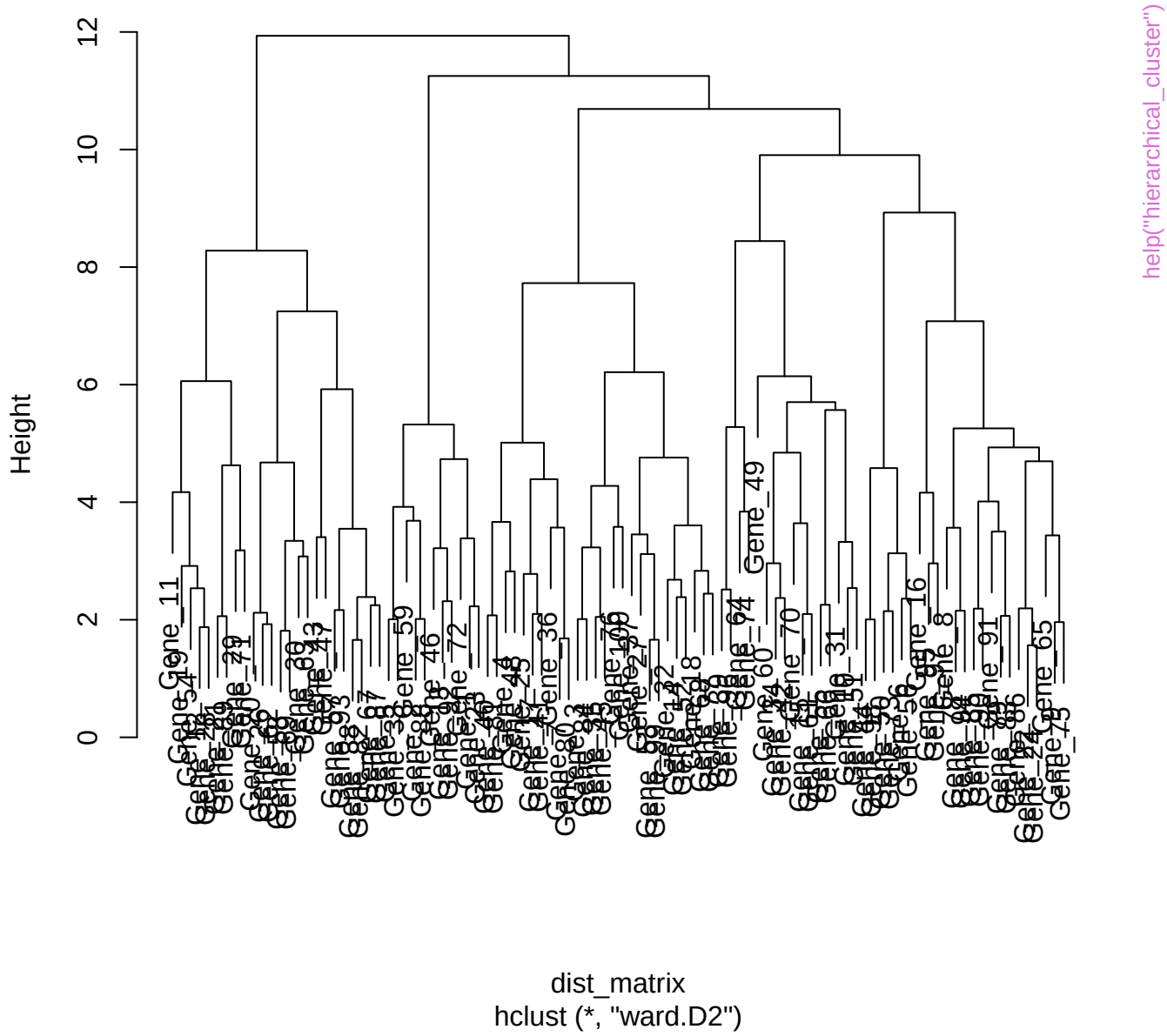
Gap Statistic for Optimal k



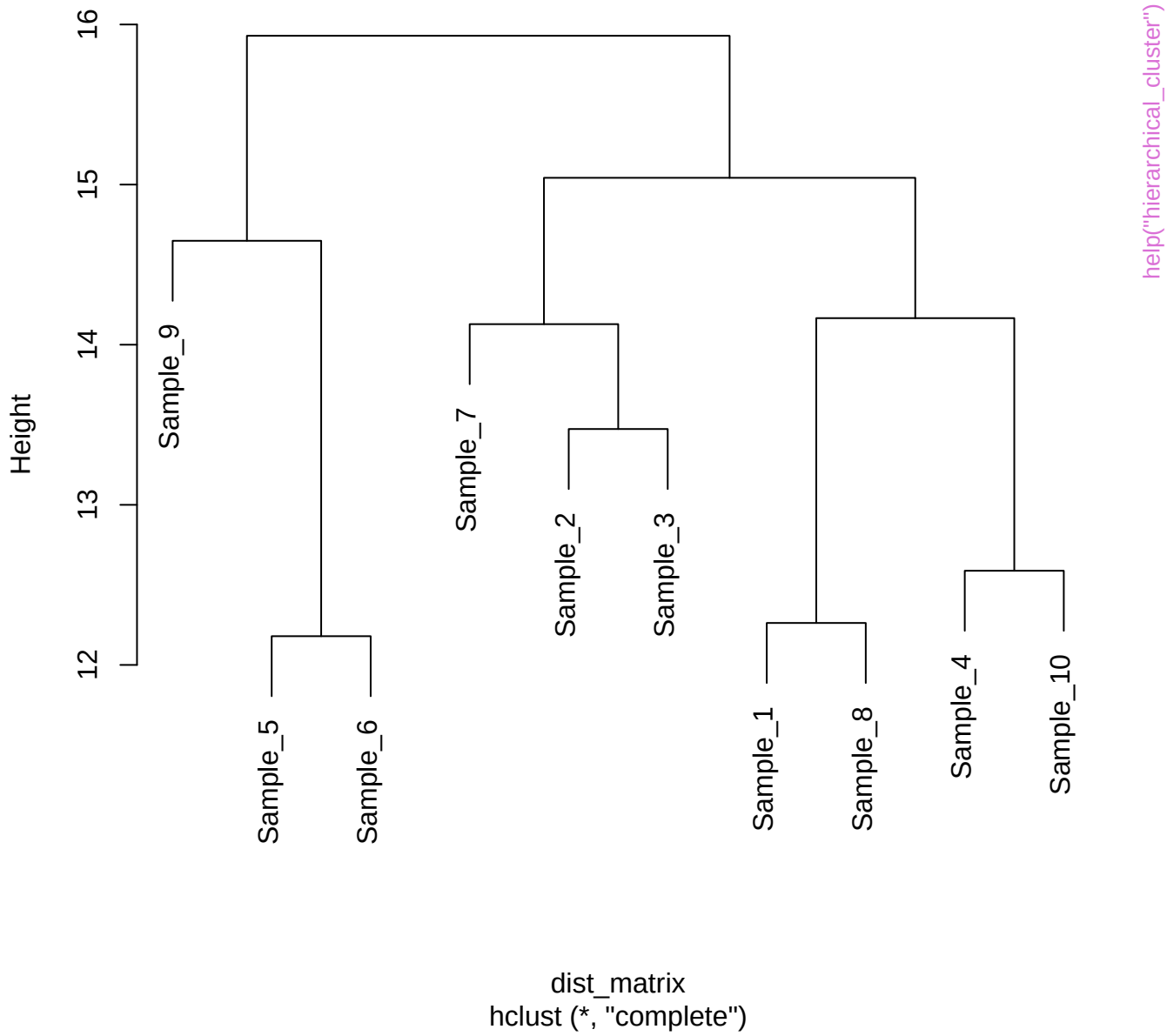
Parallel Gap Statistic Results

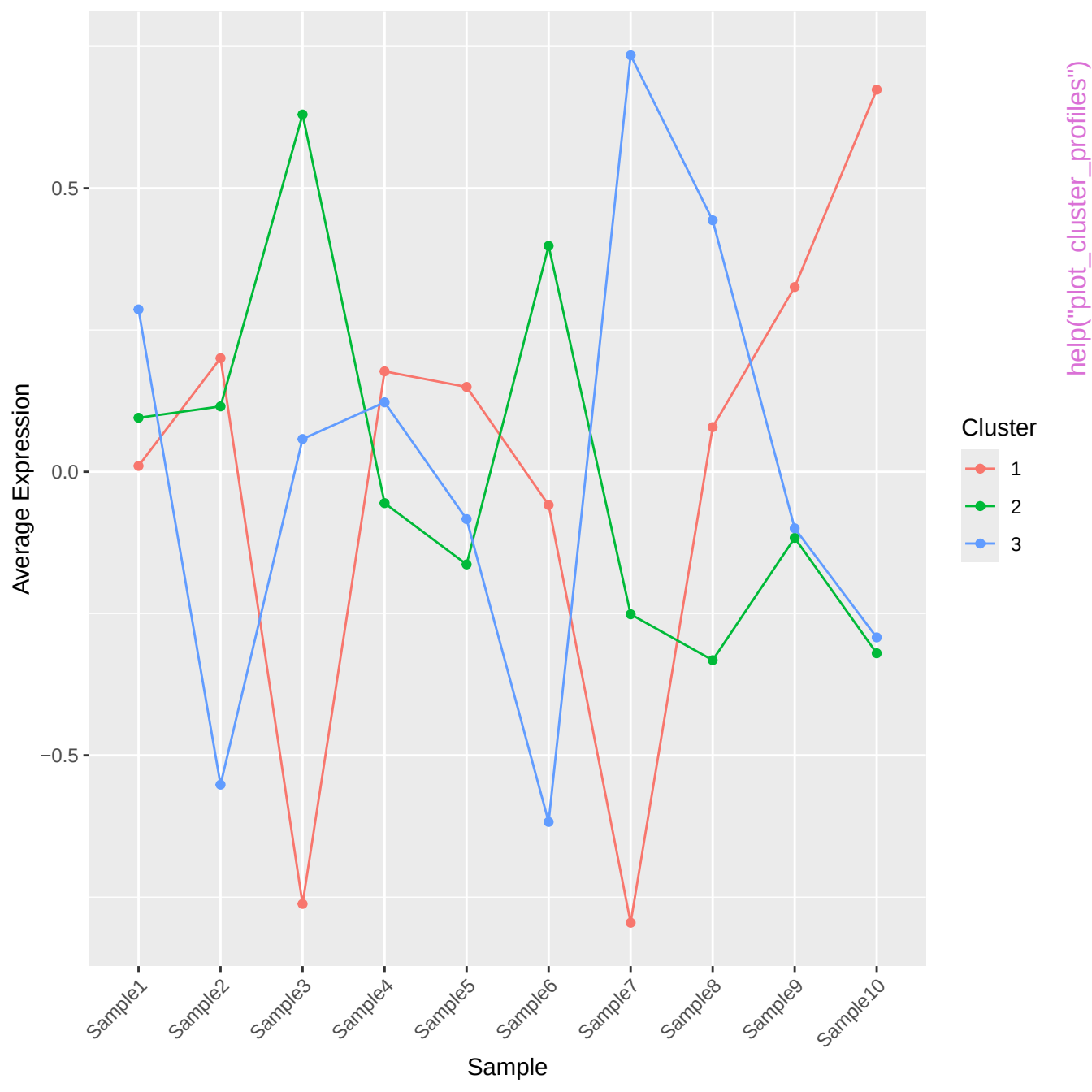


Gene Clustering

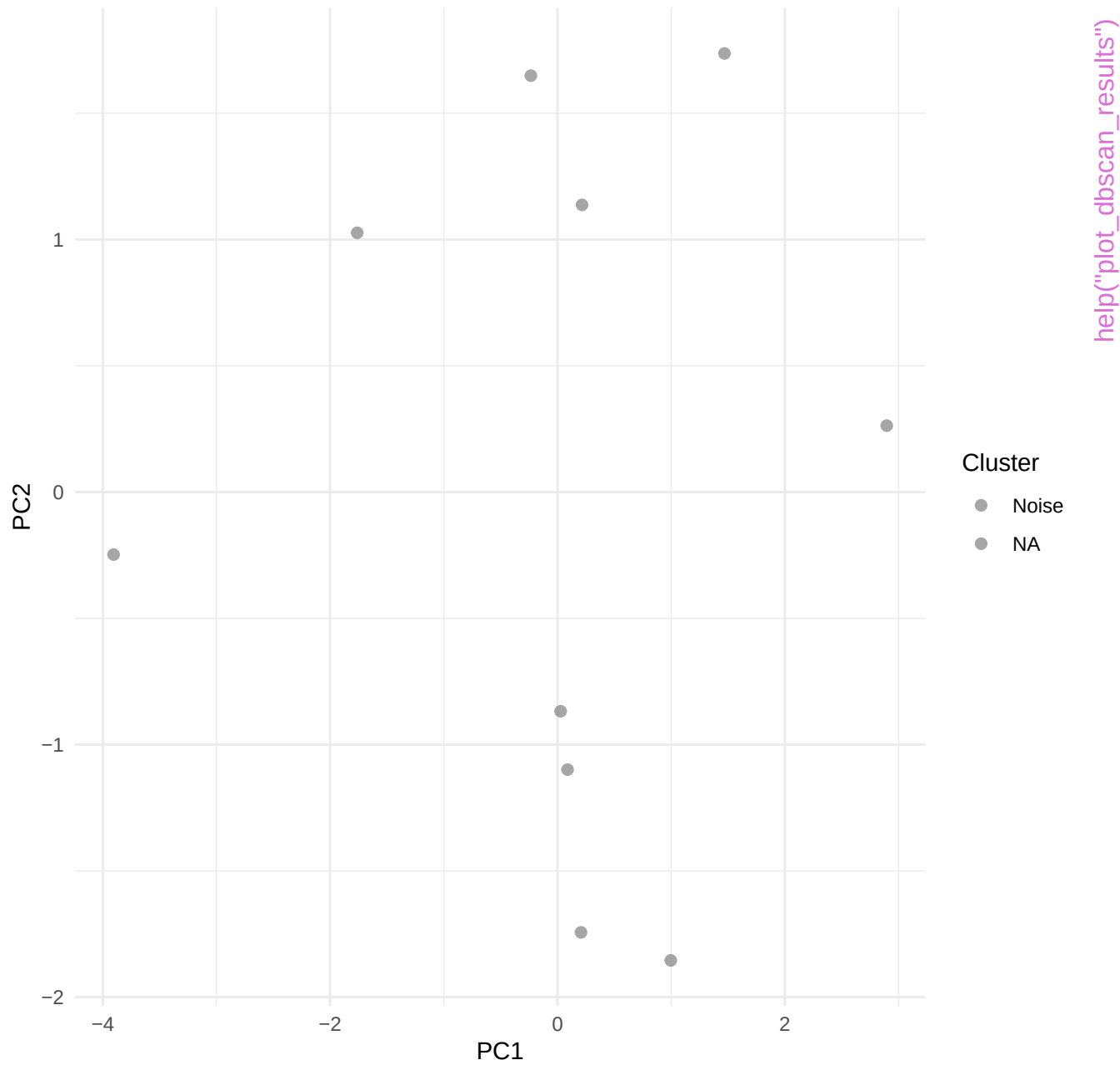


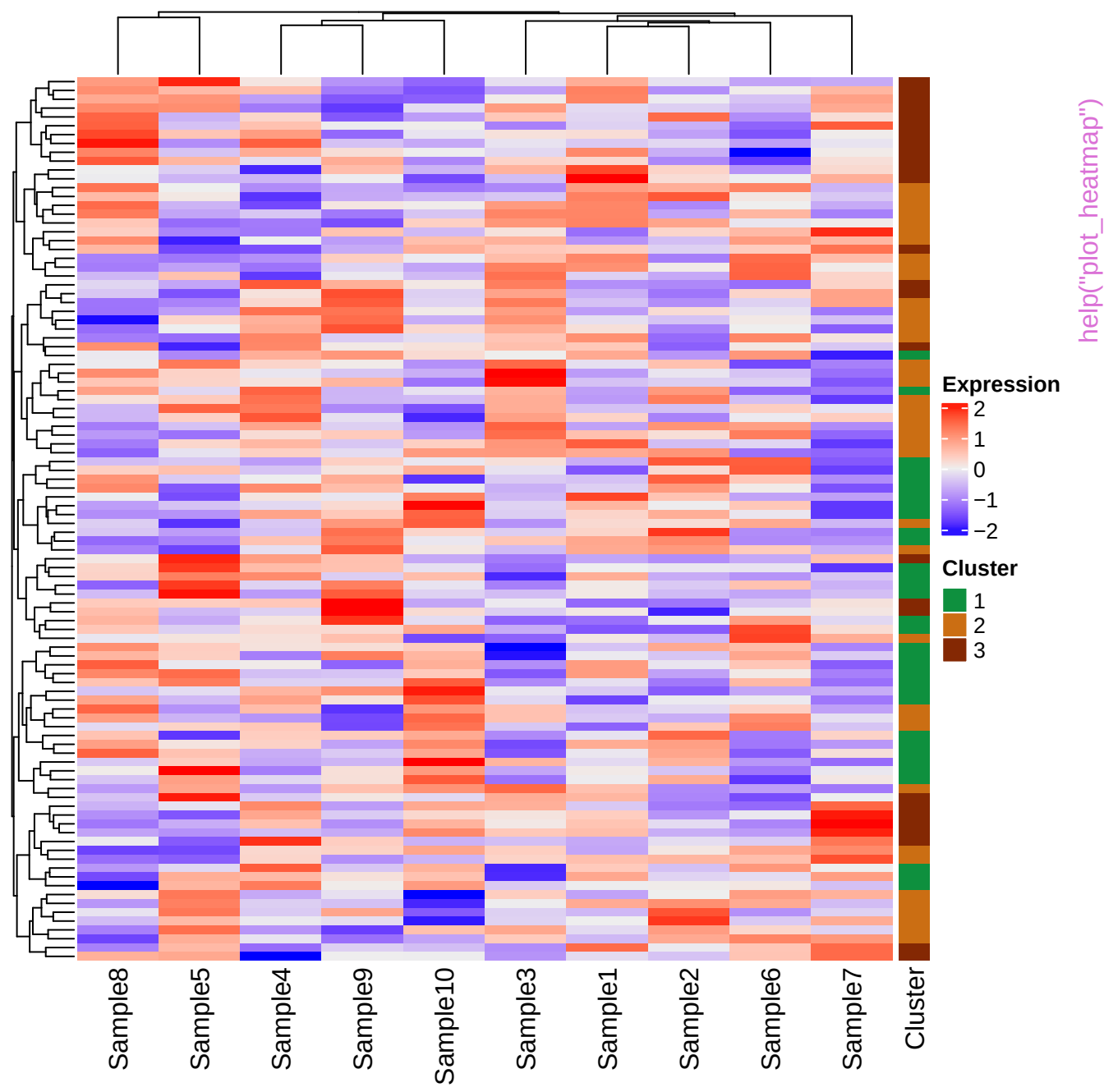
Sample Clustering



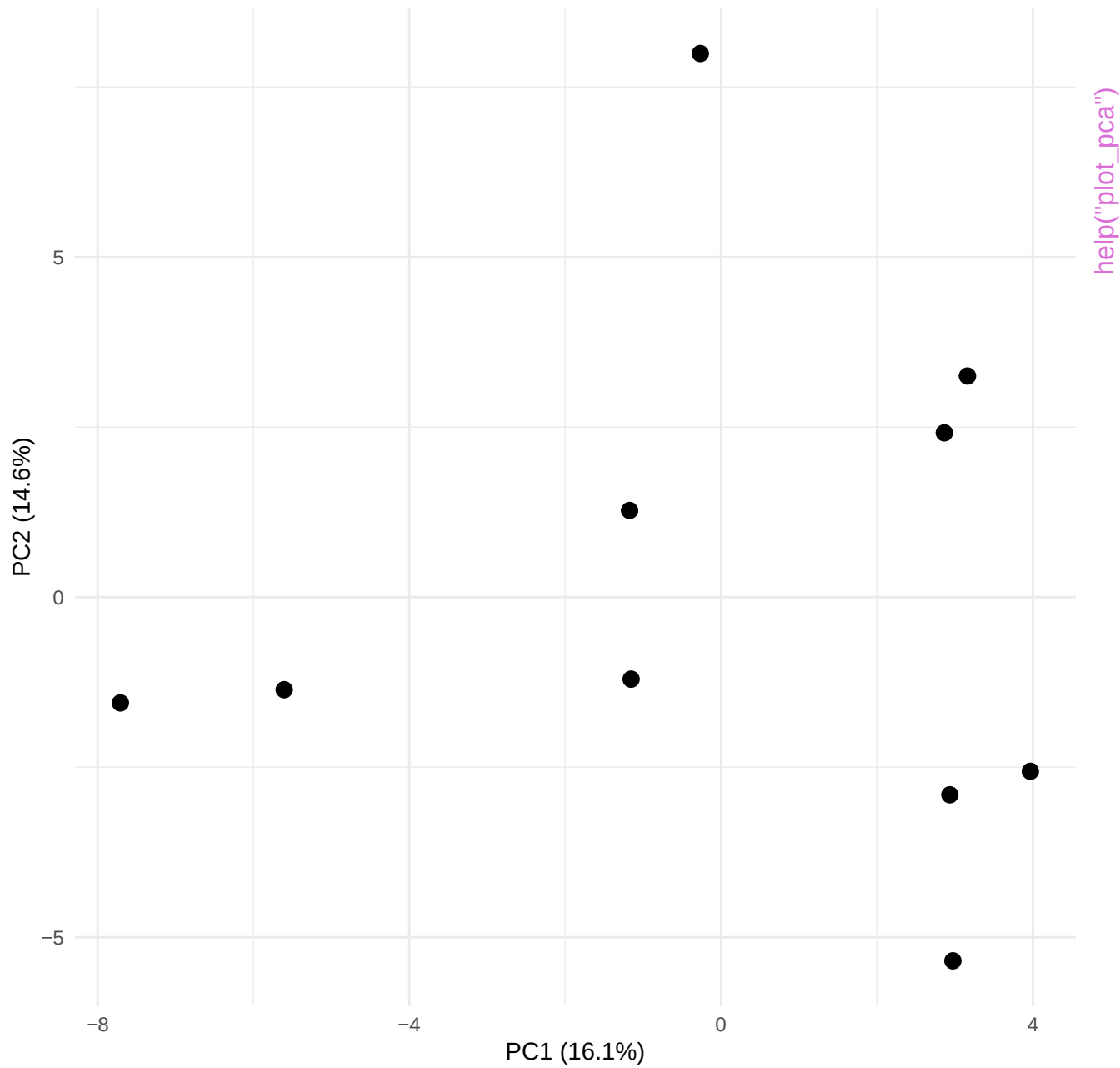


DBSCAN Clustering on gene

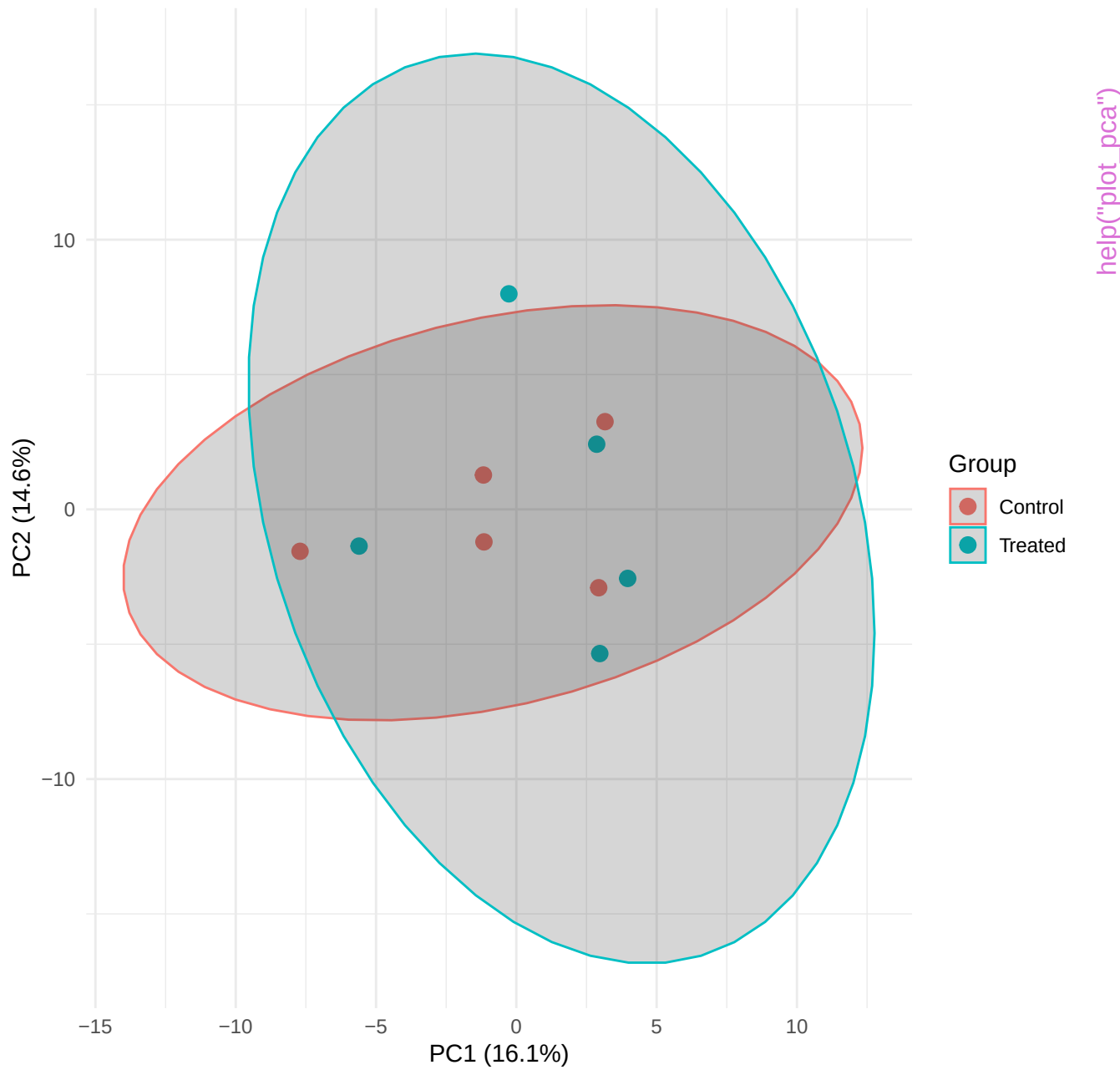




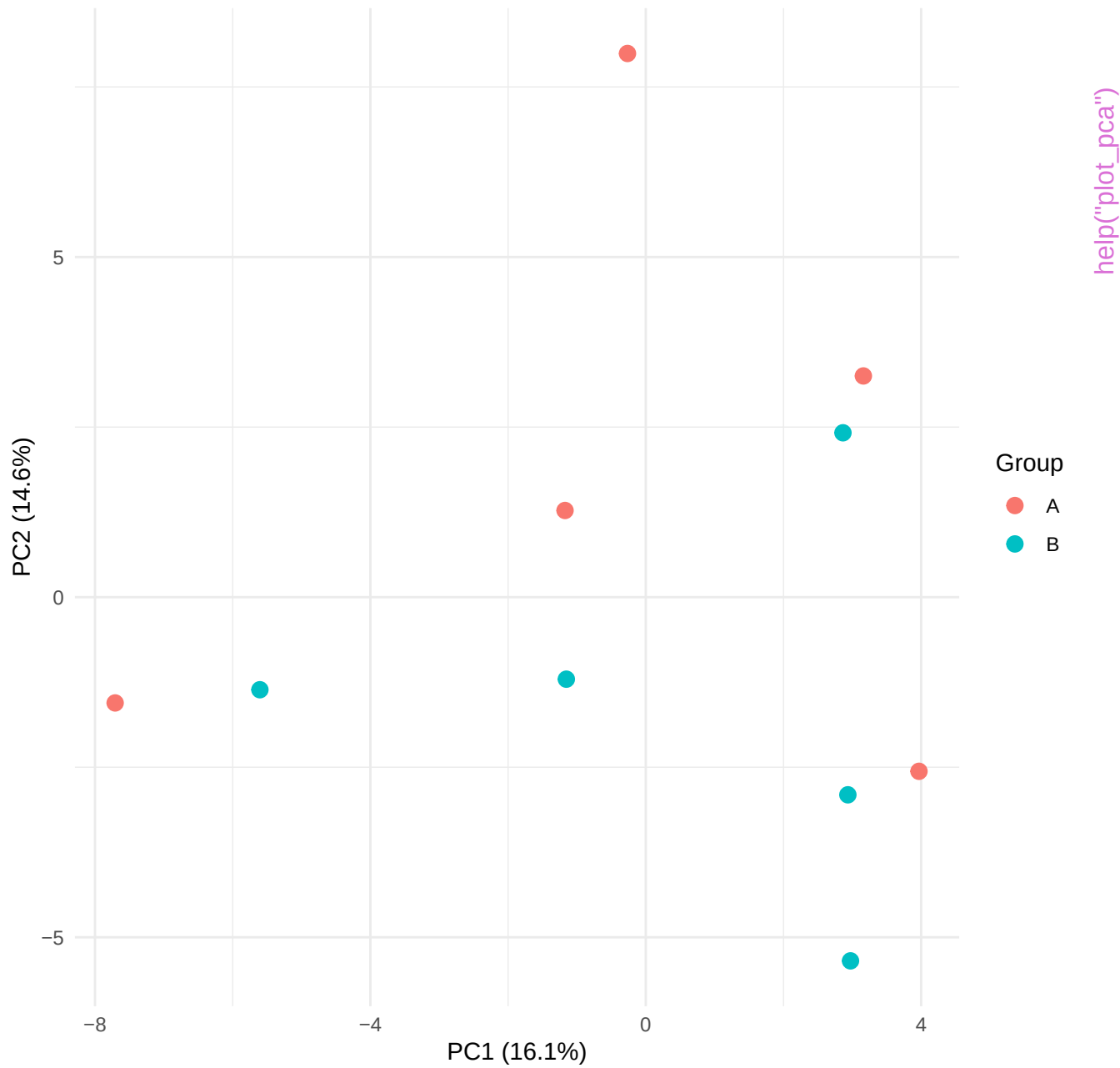
PCA Plot



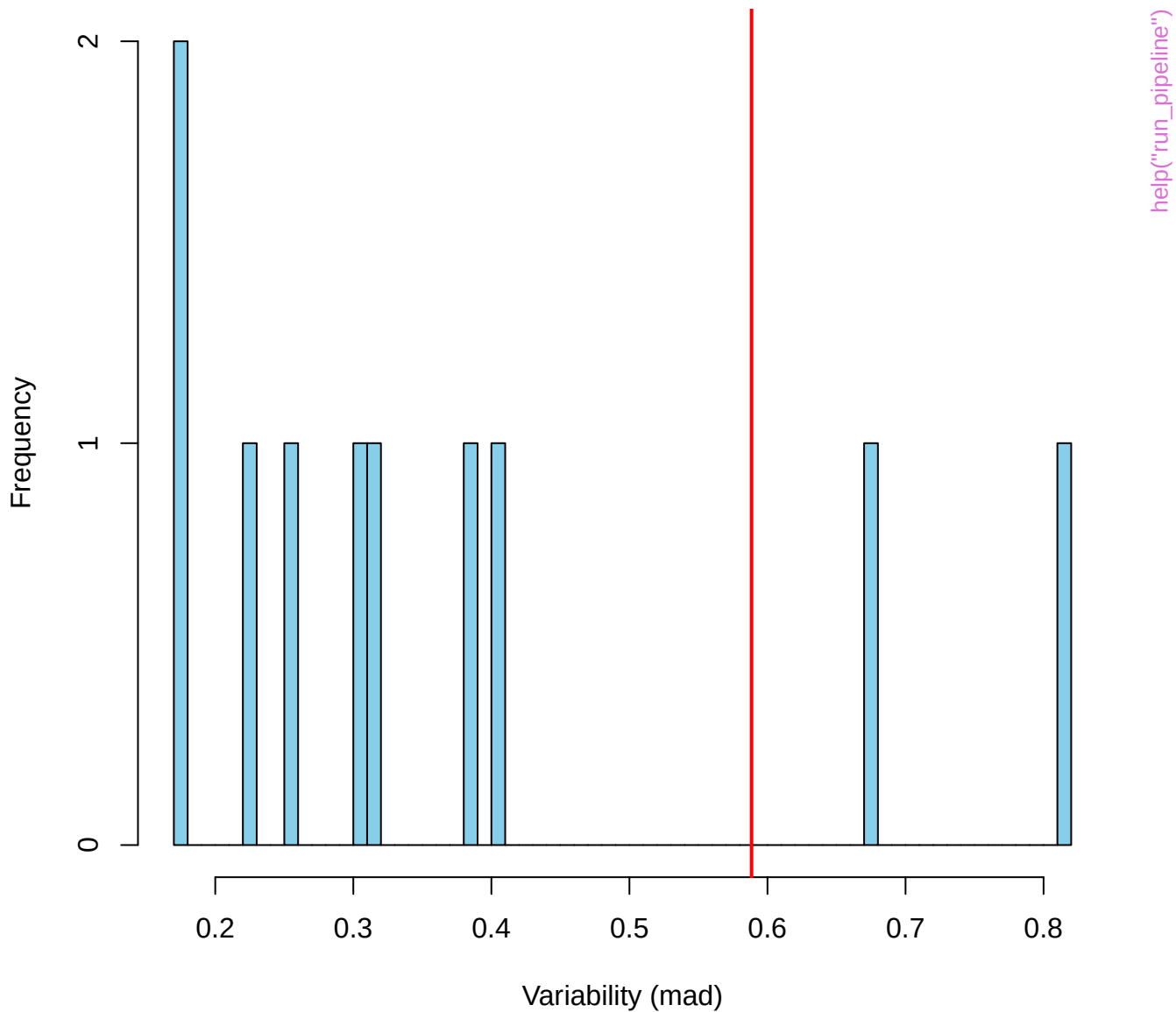
PCA Plot



PCA by Batch



Gene Variability (mad)

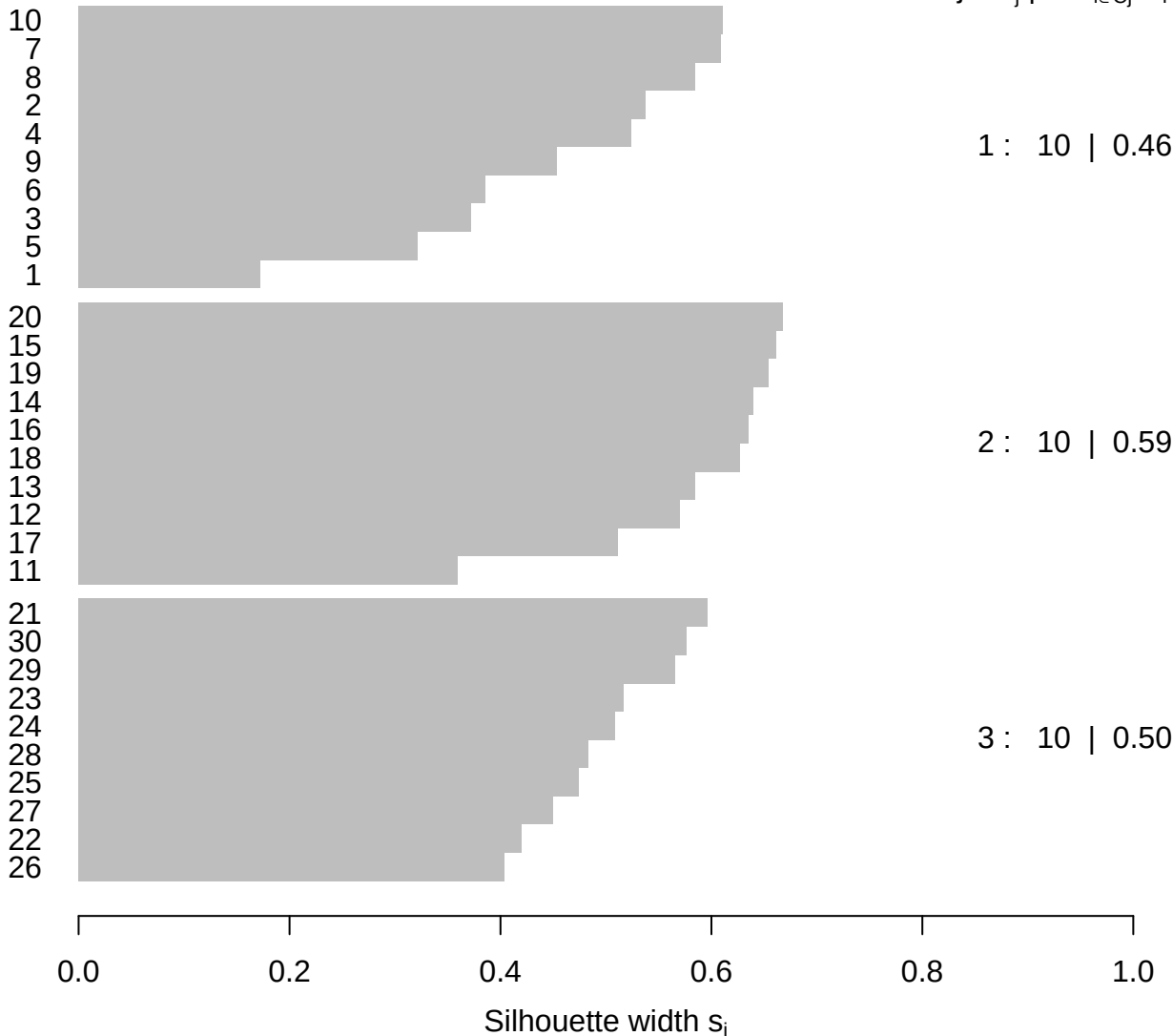


Silhouette plot of (x = clusters, dist = d)

n = 30

3 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$
[help\("silhouette score"\)](#)



Average silhouette width : 0.52